

IPS - PSP Aurora PostgreSQL

1.

- 1.
- 2. Background
- 3. Scope
- 4. Create APG Cluster
 - 4.1. Setup PSP Data Store
 - 4.2. Create shared/pre-requisite config
 - 4.3. Create Cluster Stack Configs
 - 4.4. Create Deploy.json Config
 - 4.5. Test and Deploy Shared Stack Config
 - 4.6. Test and Deploy Cluster Stack Configs
- 5. Monitoring
 - 5.1. us-west-2
 - 5.2.

2. Background

Build Aurora PostgreSQL database with monitoring and alerting as per Intuit standards using IPS automation for relational database.

3. Scope

- Review all the prerequisite, Aurora PostgreSQL cluster and instance configurations are created as specified in the JSON template
- Naming convention looks appropriate for all the resources
- Parameter add, modify works at both cluster and instance level

Monitoring and alerting works as expected

4. Create APG Cluster

4.1. Setup PSP Data Store

For local test purpose -

Refer to <https://devportal.intuit.com/app/dp/capability/3821321876122781829/capabilityDocs/docs/quick-start-guide.md> for setting up your own Data Store.

Note: Please ensure that you are not using any existing config/stack/env name.

4.2. Create shared/pre-requisite config

pre-prod: [psp-rds-prerequisites-ppd.json](#)

Below are environment specific values for all the "id" fields in the Config. Following are the five "id" fields examples from pre-prod config. Please look for each of them and replace if needed.

```
"id": "psp",  
"id" : "monolith-pg-13-cluster"  
"id" : "ibob-pg-13-Cluster"  
"id" : "monolith-pg-13-instance"  
"id" : "ibob-pg-13-instance"  
"id" : "ppddbsubnetgrp"  
"id" : "ppddbsecuritygrp"
```

4.3. Create Cluster Stack Configs

1. Cluster Configs are under the "config" folder of the above downloaded git repo .

If you are creating a new cluster then you can clone an existing file and then rename/update the cloned file with the new cluster id.

ppd : [psppsp_ibob-ppd.json](#)

2. Update environment specific values for the listed fields below. You can refer to "Naming Convention" section above.

Following are the examples from QA config. Please look for each of them and replace if needed.

```
"id": "psp"

"description": "sys ibob"

"clusterParameterGroupRef": "psp:ibob-pg-13-Cluster"

"instanceParameterGroupRef": "psp:ibob-pg-13-instance"

"subnetGroupRef": "psp:ppddbsubnetgrp"

"securityGroupRef" : "psp:ppddbsecuritygrp"

"name": "dev"

"id": "ibob"
```

Note: Please ensure **Cluster Config** is using the required **parameter/security/subnet** groups from the required **Shared Config**

4.4. Create Deploy.json Config

There is a third type of config file called deploy.json that helps with deployment structure

This file is directly under the [ips](#) folder

1. It lets you define the targets
2. For each target you can set env as well as clusters

```
{
  "targets": [
    {
      "name": "primary-cluster",
      "accountId": "1111-2222-3333",
      "vpcId": "vpc-01010101010101",
      "region": "us-west-2"
    }
  ],
  "stages": [
    {
      "name": "ppd-clusters",
      "deployments": [
        {
          "environment": "e2e",
          "group": "primary-mockclient",
          "target": "primary-cluster"
        }
      ]
    }
  ]
}
```

4.5. Test and Deploy Shared Stack Config

Commands that can be used with cdk-

list : Lists the stacks in the app

diff : Compares the specified stack with the deployed stack or a local CloudFormation template

synth : Synthesizes and prints the CloudFormation template for the specified stack(s)

deploy : Deploys the specified stack(s)

Ref- <https://docs.aws.amazon.com/cdk/latest/guide/cli.html>

NOTE: CDK needs AWS Credentials to run. Please setup/download Temporary API Key via command line on or DevPortal

Go to the IPS datastore dir on your local

```
### Onetime ONLY bootstrapping for a region

cdk bootstrap aws://<AWS ccount>/<region> --require-approval=never -c env=<env name> -c config=config/< Shared Stack Config created above >

i.e.,
cdk bootstrap aws://152430470825/us-west-2 --require-approval=never -c env=dev -c config=psp-rds-prerequisites-mockclient.json

#####

### List the stacks

cdk list -c env= <env name> -c config=config/< Shared Stack Config created above >

i.e.,

cdk list --all -c env=e2e -c group=primary-mockclient -c config=psp-rds-prerequisites-mockclient.json

#####

## For the below commands either you pick one of the listed stacks from the above "list" command or use "--all"

cdk <COMMAND> < --all or a specific stack listed by list command > -c env= <env name> -c config=< Shared Stack Config created above >

i.e.,

cdk diff --all -c env=e2e -c group=primary-mockclient -c config=psp-rds-prerequisites-mockclient.json

cdk synth pspmock-shared -c env=e2e -c group=primary-mockclient -c config=psp-rds-prerequisites-mockclient.json

cdk deploy --all -c env=e2e -c config=psp-rds-prerequisites-mockclient.json
```

Note:

1. Run `cdk diff` to make sure no issues before you deploy
2. Please ensure you are using the right stack in the `cdk` command

4.6. Test and Deploy Cluster Stack Configs

Go to the IPS datastore dir on your local (*quickstart-ds* or whatever you named it)

```
cdk list -c env= <env name> -c group= <Group name specified in cluster & deploy jsons > -c config= < Cluster Stack Config created above >
```

i.e.,

```
cdk list --all -c env=e2e -c group=primary-mockclient -c config=psp_mock-client.json
```

```
#####
```

```
## For the below commands either you pick one of the listed stacks from the above "list" command or use "--all"
```

```
cdk <COMMAND> < --all or a specific stack listed by list command > -c env= <env name> -c group= <Group name specified in cluster & deploy jsons > -c config= < Cluster Stack Config created above >
```

i.e.,

```
cdk diff --all -c env=e2e -c group=primary-mockclient -c config=psp_mock-client.json
cdk synth pspmock-shared -c env=e2e -c group=primary-mockclient -c config=psp_mock-client.json
cdk deploy --all -c env=e2e -c group=primary-mockclient -c config=psp_mock-client.json
```

Note:

1. Run `cdk diff` to make sure no issues before you deploy
2. Please ensure you are using the right stack in the `cdk` command

Upgrade from IPS 0.6.14 to 0.8.10

Please refer for new updates <https://devportal.intuit.com/app/dp/capability/3821321876122781829/capabilityDocs/main/docs/upgrading.md>

1. Update pom.xml

First update pom.xml to ips-aurora-cdk [pom.xml](#)

2. Migrate shared resources

Using the OLD common config files, do a `cdk list` (NOT a `cdk deploy`!!) on each "shared" stack. This will prompt our app to migrate resources out of each "shared" stack, and into the correct new location (either the vpc or region stack as appropriate). You've already done this for some of the resources

```
For "spsp-sys-db" cluster -- migrate resources out of the "spsp-shared" stack (and delete the stack):
cdk list --all -c env=pds -c group=primary-pds -c config=config-old/sharedStacks/psp_pre-req_sys.json
For "psp-shared-aud" cluster -- migrate resources out of the "psp-shared" stack (and delete the stack):
cdk list --all -c env=pds -c group=primary-pds -c config=config-old/sharedStacks/psp_shared_pre-req.json
After the cdk list step, all shared resources should have been relocated to one of the IpsRds- stacks, and the
"shared" stacks should have been deleted.
```

From this point on, you can ignore the files in the `config-old/` dir.

3. Deploy common stacks

After migrating all shared resources (and deleting all "shared" stacks) ... use the NEW common config files, and do a `cdk deploy` on the common IpsRds-stacks -- shared resources are all deployed together now, so one env per account/region combo should be enough:

```
cdk deploy --all -c env=pds -c config=psp-prerequisites-pds.json
```

4. Deploy clusters

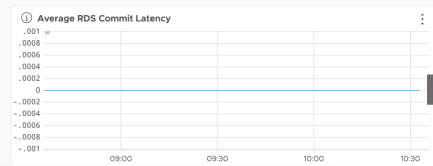
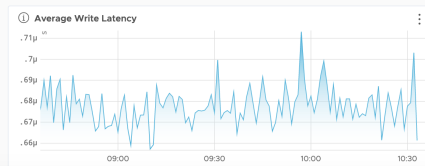
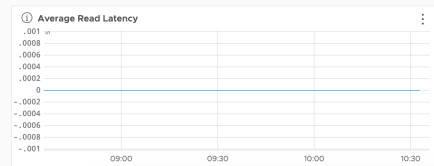
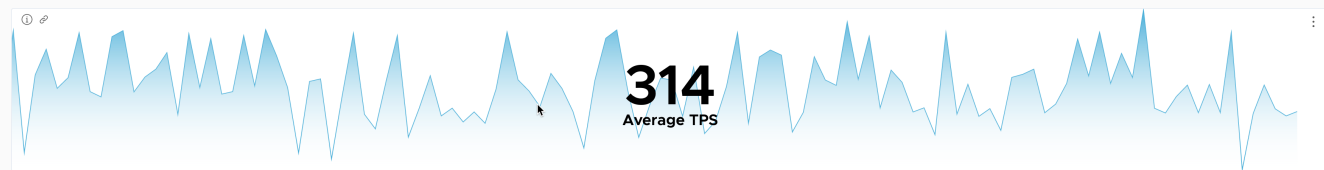
Now you should be able to deploy your clusters the normal way. Upgrade each of them to 0.8.10 and they should pull all the same shared resources, from the new location.

5. Monitoring

5.1. us-west-2

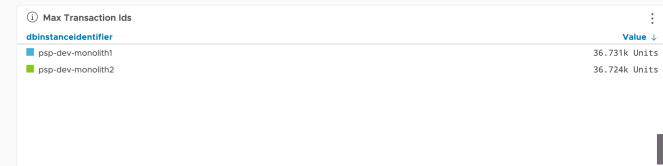
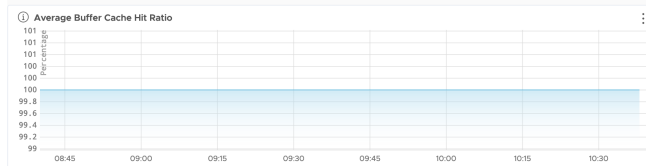
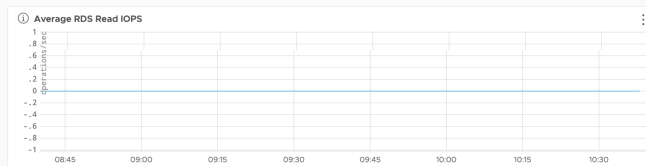
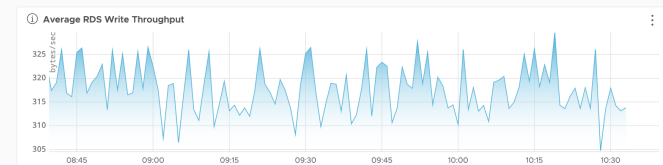
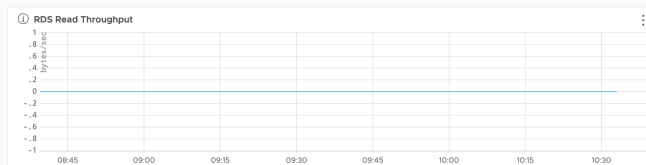
<https://intuit.wavefront.com/u/tTqc9fpr2G?t=intuit>

Golden Signals



5.2.

Query metrics

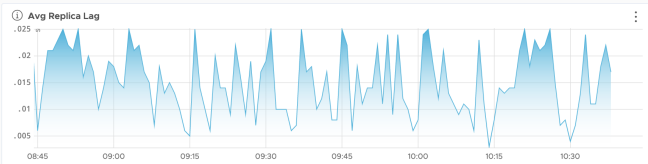
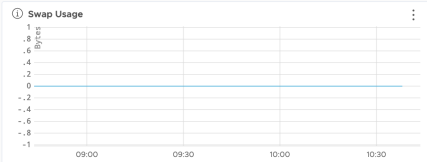
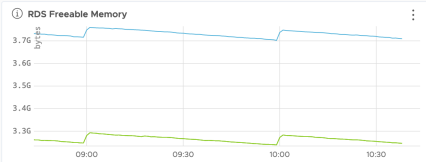


Instance Metrics

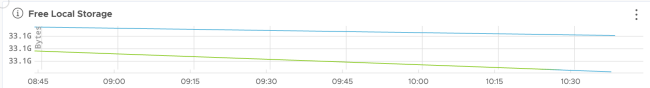
Instance Info						
db_instance_class	dbinstanceidentifier	network_performance	total_mem	vcpu	Value	
r5.large	psp-dev-monolith2	Up to 10 Gigabit	16384	2	1,000	
r5.large	psp-dev-monolith1	Up to 10 Gigabit	16384	2	1,000	

RDS Deadlocks		
dbinstanceidentifier	Value	
psp-dev-monolith1	0.000	
psp-dev-monolith2	0.000	

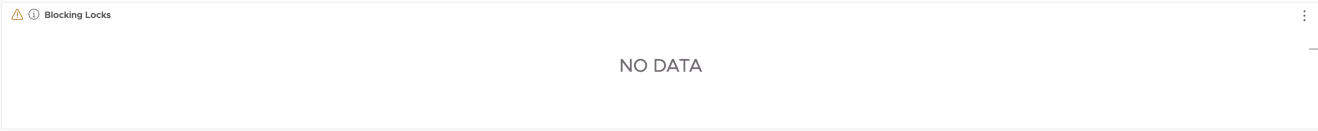
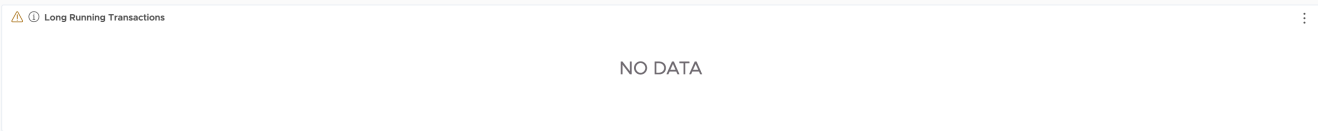
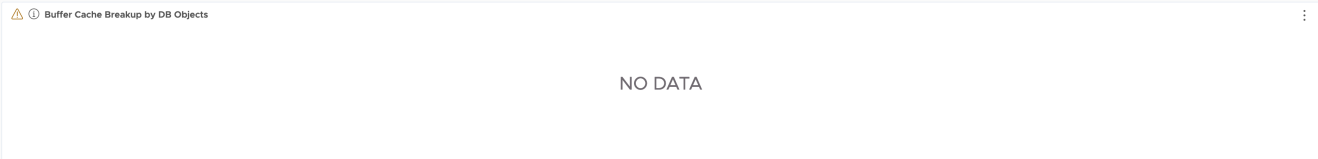
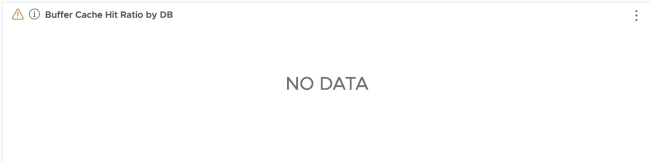
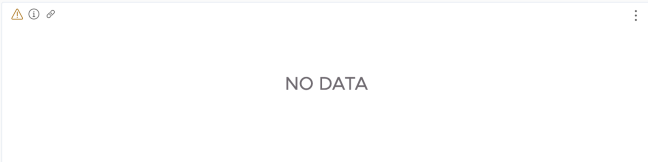
RDS Database Connections		
dbinstanceidentifier	Value	
psp-dev-monolith1	0.000	
psp-dev-monolith2	0.000	



Database Connections		
dbinstanceidentifier	Value	
psp-dev-monolith1	0.000	Connections
psp-dev-monolith2	0.000	Connections



DB Insights



▼ Growth Trends

